Weak Nullstellensatz

An End-to-End Meta-Math Walkthrough

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Abstract

A field that is finitely generated as a k-algebra cannot contain a transcendental generator: integrality would force a polynomial ring with only one denominator inverted to be a field. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AG-02 / Algebraic Geometry / Report v1

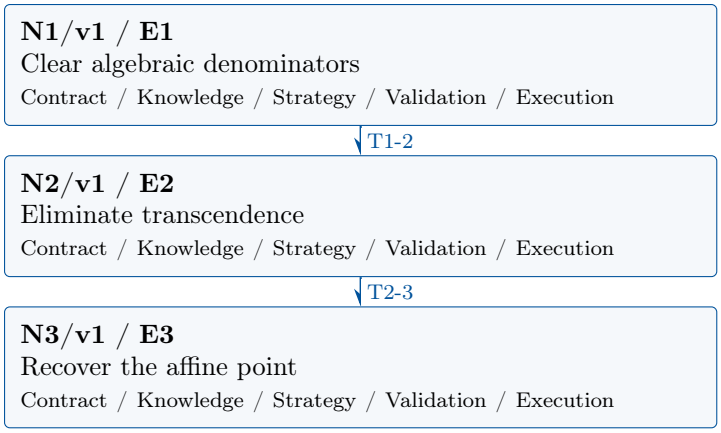
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let k be algebraically closed and $A = k[x_1, \ldots, x_n]$, with finite $n \geq 0$. Every maximal ideal is $(x_1 - a_1, \ldots, x_n - a_n)$ for a unique point $a \in k^n$. Consequently every proper ideal of A has a common zero in k^n.*

A. Foundations and permitted imports

Use existence of maximal ideals containing proper ideals (Zorn's lemma), finite transcendence bases chosen from a finite generating set, and elementary field algebra. Import the integrality facts: finitely many integral elements generate a finite module; every element of a finite-module algebra is integral, by the determinant trick. An element is integral over a ring if it satisfies a monic polynomial with coefficients in that ring. The finite-generation field lemma needed below is proved here, not imported as the target.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. For algebraically independent $t_1, \ldots, t_r$, there is nonzero $d \in k[t_1, \ldots, t_r]$ such that K is integral over $B = k[t_1, \ldots, t_r, 1/d]$.

Receiving activity. N2.1

Compatibility. E1/v1, with K a field integral over the domain B. Check: The field k is infinite; choose c so that $t_1 - c$ does not divide d.

T2-3: N2/E2/v1 to N3. $r = 0$, and hence $K = k$ for every finitely generated k-algebra that is a field.

Receiving activity. N3.1

Compatibility. A maximal ideal $\mathfrak{m} \subset A$ and E2/v1 applied to $K = A/\mathfrak{m}$. Check: The quotient is a field because the ideal is maximal; the scalar identification must respect the given embedding of k.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Clear algebraic denominators

Object and role. Separate a maximal algebraically independent subset and make the remaining generators integral after one localization. **Local obligation:** For algebraically independent $t_1, \dots, t_r$, there is nonzero $d \in k[t_1, \dots, t_r]$ such that K is integral over $B = k[t_1, \dots, t_r, 1/d]$.

Inputs. A field $K = k[b_1, \dots, b_m]$ over the algebraically closed field k.

C — Contract

Publish E1/v1 only after establishing this obligation: For algebraically independent $t_1, \dots, t_r$, there is nonzero $d \in k[t_1, \dots, t_r]$ such that K is integral over $B = k[t_1, \dots, t_r, 1/d]$.

K — Knowledge

A field $K = k[b_1, \dots, b_m]$ over the algebraically closed field k. Retain the stated assumptions and import versions.

S — Strategy

Separate a maximal algebraically independent subset and make the remaining generators integral after one localization.

V — Validation

Finite generation is as an algebra, not merely as a field; only finitely many denominators need to be inverted.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Choose a maximal algebraically independent subset $t_1, \dots, t_r$ of the generators. Every remaining generator u_j is algebraic over $F = k(t_1, \dots, t_r)$: otherwise the subset could be enlarged. Each u_j satisfies a monic polynomial over F.

N1.2 Let $d \neq 0$ be the product of the finitely many polynomial denominators in these monic equations (take $d = 1$ if there are none). The u_j are integral over $B = k[t_1, \dots, t_r, 1/d]$. Because K is a field, $1/d \in K$; because $K = k[t_1, \dots, t_r, u_1, \dots, u_s]$ as an algebra, $B[u_1, \dots, u_s] = K$. The stated integrality facts now show K is integral over B.

Export E1/v1. For algebraically independent $t_1, \dots, t_r$, there is nonzero $d \in k[t_1, \dots, t_r]$ such that K is integral over $B = k[t_1, \dots, t_r, 1/d]$.

Acceptance gate. Finite generation is as an algebra, not merely as a field; only finitely many denominators need to be inverted.

Targeted repair. If arbitrary rational coefficients are treated as polynomial coefficients, explicitly collect their finitely many denominators into d. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Eliminate transcendence

Object and role. First force B to be a field, then find a polynomial that cannot become a unit in that localization. **Local obligation:** $r = 0$, and hence $K = k$ for every finitely generated k-algebra that is a field.

Inputs. E1/v1, with K a field integral over the domain B. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $r = 0$, and hence $K = k$ for every finitely generated k-algebra that is a field.

K — Knowledge

E1/v1, with K a field integral over the domain B. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

First force B to be a field, then find a polynomial that cannot become a unit in that localization.

V — Validation

The field k is infinite; choose c so that $t_1 - c$ does not divide d.

E — Execution

Carry out activities N2.1–N2.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For $0 \neq b \in B$, the element $b^{-1} \in K$ is integral: $(b^{-1})^q + c_{q-1}(b^{-1})^{q-1} + \cdots + c_0 = 0$. Multiplying by b^{q-1} expresses b^{-1} as an element of B. Thus B is a field.

N2.2 If $r \geq 1$, choose $c \in k$ with $t_1 - c \nmid d$. Such a c exists because a nonzero polynomial has only finitely many distinct factors $t_1 - c$, whereas an algebraically closed field is infinite. If $t_1 - c$ were invertible in B, then $(t_1 - c)q = d^s$ for some polynomial q and integer $s \geq 0$. Substituting $t_1 = c$ gives a contradiction since $d(c, t_2, \ldots, t_r) \neq 0$. Thus B would not be a field. Therefore $r = 0$.

N2.3 All generators are now algebraic over k, and finitely many algebraic generators give a finite algebraic extension. Algebraic closedness implies that extension is k itself. This proves the required finite-generation field lemma for this base field.

Export E2/v1. $r = 0$, and hence $K = k$ for every finitely generated k-algebra that is a field.

Acceptance gate. The field k is infinite; choose c so that $t_1 - c$ does not divide d.

Targeted repair. If the localization is declared a field without proof, derive inverses from monic equations; if transcendence is ignored, use the nonunit $t_1 - c$ witness. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Recover the affine point

Object and role. Identify each coordinate residue with a scalar and identify the kernel of evaluation.
Local obligation: Maximal ideals are exactly point ideals, and every proper ideal has a common zero.

Inputs. A maximal ideal $\mathfrak{m} \subset A$ and E2/v1 applied to $K = A/\mathfrak{m}$. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: Maximal ideals are exactly point ideals, and every proper ideal has a common zero.

K — Knowledge

A maximal ideal $\mathfrak{m} \subset A$ and E2/v1 applied to $K = A/\mathfrak{m}$. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Identify each coordinate residue with a scalar and identify the kernel of evaluation.

V — Validation

The quotient is a field because the ideal is maximal; the scalar identification must respect the given embedding of k.

E — Execution

Carry out activities N3.1–N3.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 The field $A/\mathfrak{m}$ is generated as a k-algebra by the coordinate residues. E2 identifies it with k as a k-algebra. Write the residue of x_i as a_i. Then every $x_i - a_i \in \mathfrak{m}$.

N3.2 Evaluation at a has kernel $(x_1 - a_1, \ldots, x_n - a_n)$: repeatedly replace coordinates by their values, so $f(x) - f(a)$ is a sum of multiples of $x_i - a_i$. That point ideal is maximal, since evaluation has quotient k, and inclusion forces equality with $\mathfrak{m}$. Uniqueness follows by comparing coordinate values.

N3.3 If I is proper, place it inside a maximal ideal and use its point a. Every polynomial in I vanishes at a. For $n = 0$, the only proper ideal of k is zero, and the unique empty coordinate tuple gives the same statement.

Export E3/v1. Maximal ideals are exactly point ideals, and every proper ideal has a common zero.

Acceptance gate. The quotient is a field because the ideal is maximal; the scalar identification must respect the given embedding of k.

Targeted repair. If a nonmaximal ideal is treated as having a field quotient, first pass to a maximal ideal containing it. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Clear algebraic denominators

1. **N1.1.** Choose a maximal algebraically independent subset $t_1, \ldots, t_r$ of the generators. Every remaining generator u_j is algebraic over $F = k(t_1, \ldots, t_r)$: otherwise the subset could be enlarged. Each u_j satisfies a monic polynomial over F.

2. **N1.2.** Let $d \neq 0$ be the product of the finitely many polynomial denominators in these monic equations (take $d = 1$ if there are none). The u_j are integral over $B = k[t_1, \ldots, t_r, 1/d]$. Because K is a field, $1/d \in K$; because $K = k[t_1, \ldots, t_r, u_1, \ldots, u_s]$ as an algebra, $B[u_1, \ldots, u_s] = K$. The stated integrality facts now show K is integral over B.

B. N2 — Eliminate transcendence

1. **N2.1.** For $0 \neq b \in B$, the element $b^{-1} \in K$ is integral: $(b^{-1})^q + c_{q-1}(b^{-1})^{q-1} + \cdots + c_0 = 0$. Multiplying by b^{q-1} expresses b^{-1} as an element of B. Thus B is a field.

2. **N2.2.** If $r \geq 1$, choose $c \in k$ with $t_1 - c \nmid d$. Such a c exists because a nonzero polynomial has only finitely many distinct factors $t_1 - c$, whereas an algebraically closed field is infinite. If $t_1 - c$ were invertible in B, then $(t_1 - c)q = d^s$ for some polynomial q and integer $s \geq 0$. Substituting $t_1 = c$ gives a contradiction since $d(c, t_2, \ldots, t_r) \neq 0$. Thus B would not be a field. Therefore $r = 0$.

3. **N2.3.** All generators are now algebraic over k, and finitely many algebraic generators give a finite algebraic extension. Algebraic closedness implies that extension is k itself. This proves the required finite-generation field lemma for this base field.

C. N3 — Recover the affine point

1. **N3.1.** The field $A/\mathfrak{m}$ is generated as a k-algebra by the coordinate residues. E2 identifies it with k as a k-algebra. Write the residue of x_i as a_i. Then every $x_i - a_i \in \mathfrak{m}$.

2. **N3.2.** Evaluation at a has kernel $(x_1 - a_1, \ldots, x_n - a_n)$: repeatedly replace coordinates by their values, so $f(x) - f(a)$ is a sum of multiples of $x_i - a_i$. That point ideal is maximal, since evaluation has quotient k, and inclusion forces equality with $\mathfrak{m}$. Uniqueness follows by comparing coordinate values.

3. **N3.3.** If I is proper, place it inside a maximal ideal and use its point a. Every polynomial in I vanishes at a. For $n = 0$, the only proper ideal of k is zero, and the unique empty coordinate tuple gives the same statement.

□

V. WORKED ORCHESTRATION

Over $\mathbb{C}$, take $I = (x^2 + 1, y - x)$. The points (i, i) and $(-i, -i)$ both annihilate I. A maximal ideal above I can be $(x - i, y - i)$; the theorem guarantees a point for each maximal ideal, not uniqueness of the common zero of an arbitrary proper ideal.

A. Boundary and hypothesis audit

Over $\mathbb{R}$, the ideal $(x^2 + 1)$ is maximal but has no real zero. Its quotient is $\mathbb{C}$, a proper finite algebraic extension of $\mathbb{R}$. This pinpoints the algebraic-closure assumption.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 claim: a field finitely generated as a field extension must be algebraic. The field $k(t)$ is generated by t as a field and is transcendental. **Diagnosis:** Knowledge confused field generation with algebra generation. **Repair:** use the actual quotient presentation as a finitely generated algebra in N1; it is precisely what yields $K = B[u_1, \ldots, u_s]$. No change to the theorem is needed. Recheck E1, E2, and the maximal-ideal identification.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use the maximal-ideal result in AG-03 to recover the full vanishing ideal by the Rabinowitsch construction.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. J. S. Milne, Algebraic Geometry, chapters on commutative algebra, algebraic sets, and affine varieties; author's mathematics site.